GroupMemory: reuse by exact symmetry

A fixed ring-circuit example: equivariant forward values, invariant shared gradients.

1. Model and observables

For an even ring of n sites, $x \in \mathbb{R}^n$ and shared weights $\theta = (\beta, \alpha)$, define

$$U_\theta(x) = \underbrace{\bigotimes_{j=0}^{n-1} R_y(0.47)}_{B} \underbrace{\bigotimes_{j=0}^{n-1} R_x(\alpha)}_{A_\alpha} \underbrace{\prod_{j=0}^{n-1} R_{ZZ}^{(j,j+1 \bmod n)}(\beta)}_{E_\beta} \underbrace{\bigotimes_{j=0}^{n-1} R_y(x_j + 0.31)}_{D(x)}, \tag{1}$$

$$R_a(t) = e^{-it\sigma_a/2}, \qquad R_{ZZ}^{(j,k)}(t) = e^{-itZ_jZ_k/2}. \tag{2}$$

Products act from right to left. The ZZ factors commute. With $|\psi_\theta(x)\rangle = U_\theta(x)|0\rangle^{\otimes n}$, the requested outputs are

$$f_\theta(x)_j = \langle\psi_\theta(x)|Z_j|\psi_\theta(x)\rangle, \qquad m_\theta(x) = \frac{1}{n}\sum_{j=0}^{n-1} f_\theta(x)_j, \qquad J_\theta(x) = \nabla_\theta m_\theta(x) \in \mathbb{R}^2. \tag{3}$$

2. The symmetry supplies the reuse certificate

Let $\mathcal{G} = D_n = \langle r, s \mid r^n = s^2 = e,\ srs = r^{-1}\rangle$, with $|D_n| = 2n$. For a site permutation g, write

$$(P_g x)_j = x_{g^{-1}(j)}, \qquad \Pi_g|b_0 \cdots b_{n-1}\rangle = |b_{g^{-1}(0)} \cdots b_{g^{-1}(n-1)}\rangle. \tag{4}$$

Rotations and reflections preserve the ring's edges. Uniform local gates and shared couplings therefore give

$$\Pi_g|0\rangle^{\otimes n} = |0\rangle^{\otimes n}, \qquad \Pi_g B \Pi_g^\dagger = B, \quad \Pi_g A_\alpha \Pi_g^\dagger = A_\alpha, \quad \Pi_g E_\beta \Pi_g^\dagger = E_\beta, \tag{5}$$

$$D(P_g x) = \Pi_g D(x) \Pi_g^\dagger \quad \Longrightarrow \quad \boxed{U_\theta(P_g x) = \Pi_g U_\theta(x) \Pi_g^\dagger}. \tag{6}$$

Using $\Pi_g^\dagger Z_j \Pi_g = Z_{g^{-1}(j)}$ in (3) yields

$$\boxed{f_\theta(P_g x) = P_g f_\theta(x)}, \qquad \boxed{m_\theta(P_g x) = m_\theta(x)}, \qquad \boxed{J_\theta(P_g x) = J_\theta(x)}. \tag{7}$$

The vector output is *equivariant*; its mean and shared-weight gradient are *invariant*. The last equality follows by differentiating an identity valid for every θ.

3. Shared parameters require shared-parameter derivatives

Let $\widetilde{m}(x; \boldsymbol{\beta}, \boldsymbol{\alpha})$ allow independent edge and site angles. Then

$$\frac{\partial m}{\partial \beta} = \sum_{j=0}^{n-1} \frac{\partial \widetilde{m}}{\partial \beta_j}\bigg|_{\beta_j=\beta,\, \alpha_j=\alpha}, \qquad \frac{\partial m}{\partial \alpha} = \sum_{j=0}^{n-1} \frac{\partial \widetilde{m}}{\partial \alpha_j}\bigg|_{\beta_j=\beta,\, \alpha_j=\alpha}. \tag{8}$$

A single-gate parameter-shift rule cannot generally be applied by shifting all shared occurrences at once. The CPU demo uses reverse-mode differentiation; finite differences of the complete circuit independently check these two derivatives.

4. Canonical representatives and reconstruction

For the finite orbit $\mathcal{G}x = \{P_g x : g \in \mathcal{G}\}$, choose a deterministic representative

$$h(x) \in \arg\min_{g\in\mathcal{G}} \text{bytes}(P_g x), \qquad c(x) = P_{h(x)} x. \tag{9}$$

The minimum is lexicographic over exact float64 byte strings, with deterministic tie breaking. It is a representation convention, not approximate numerical matching. Store one forward vector and, when requested, one gradient per representative:

$$\mathcal{C}[K_\theta(x)] = (f_\theta(c(x)),\ J_\theta(c(x)) \text{ or } \varnothing), \qquad K_\theta(x) = (\sigma, \text{bytes}(\theta), \text{bytes}(c(x))). \tag{10}$$

Here σ identifies the fixed circuit, observable and numerical execution policy. In the implementation, this context is bound to an engine instance. Recover each requested result by

$$\boxed{f_\theta(x) = P_{h(x)}^{-1} \mathcal{C}[K_\theta(x)]_f}, \qquad \boxed{J_\theta(x) = \mathcal{C}[K_\theta(x)]_J}. \tag{11}$$

Different canonicalizing permutations at a tie give the same mathematical result: their relative permutation lies in the stabilizer of $c(x)$, which also fixes $f_\theta(c(x))$ by (7).

5. What is reusable across training?

At a fixed weight value, an already-cached orbit avoids new computation. A gradient request can fill a missing J while retaining the cached forward vector. A weight change, even one representable float64 step, changes the numerical problem:

$$\text{bytes}(\theta') \neq \text{bytes}(\theta) \quad \Longrightarrow \quad \text{invalidate the old per-weight result cache.} \tag{12}$$

Thus reuse applies within equal-weight requests and their exact symmetry orbits; it does not reuse arbitrary stale gradients across parameter updates.

For N cold, distinct inputs after ordinary exact-input deduplication, partitioned into M symmetry orbits at the same θ,

$$N_{\text{forward}} : N \longrightarrow M, \qquad N_{\text{gradient}} : N \longrightarrow M, \qquad \text{avoided fraction} = 1 - \frac{M}{N}. \tag{13}$$

Four distinct augmented views per base give $4 \to 1$ and 75% fewer instances. The full orbit size is

$$|\mathcal{G}x| = \frac{2n}{|\,\text{Stab}(x)|} \leq 2n; \qquad |\mathcal{G}x| = 2n \text{ only for a free orbit.} \tag{14}$$

These are counts of avoided evaluations, not a guaranteed reduction of gates inside each compiled circuit. For equal per-forward CNOT count c_{CX}, the forward-work proxy changes from Nc_{CX} to Mc_{CX}. CPU reverse-gradient instances are not QPU circuits.

6. The performance condition and the comparison that matters

For a fixed model and weight value, let T_{eval} denote one representative evaluation and T_{reuse} include canonicalization, lookup and reconstruction. Then

$$T_{\text{reuse}} < (N - M)T_{\text{eval}} \quad \Longleftrightarrow \quad \text{reuse amortizes its own overhead,} \qquad S = \frac{T_{\text{setup}} + NT_{\text{eval}}}{T_{\text{setup}} + T_{\text{reuse}} + MT_{\text{eval}}}. \tag{15}$$

End-to-end learning also charges data preparation, compilation, classical fitting, prediction and reporting. An equally symmetry-aware Qiskit implementation can obtain the same $N \to M$ reduction; a speed advantage over that baseline must be measured, not inferred from (13).

Scope

The supplied D_n certificate applies to (1), shared weights, the stated observables and ideal execution. Position-dependent gains generally break it, so the demo disables orbit reuse and retains ordinary exact-input caching. Independent statevector and gradient tests check numerical correctness. This method does not discover arbitrary circuit symmetries or establish a quantum hardware advantage.